From Narrative Inflation to Verifiable Self-Correction: An Empirical Longitudinal Case Study of Multi-Turn LLM Reasoning on the Riemann Hypothesis

從敘事膨脹到可驗證自我修正：大語言模型在長程多輪前沿數學推理中的失敗模式與修正機制實證研究

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Platform: Google Antigravity (Gemini 3.7 Pro/Flash Engine) & Perplexity Pro (SymPy Sandbox Auditor) • August 2026 • Open Science Preprint

Abstract—Current benchmarks for evaluating mathematical reasoning in Large Language Models (LLMs) predominantly focus on static, single-turn, closed-form contest problems with known ground truths. Consequently, they fail to capture how frontier AI systems behave when deployed in open-ended, long-horizon frontier scientific research spanning hundreds of iterative dialog rounds.

In this paper, we present an extensive empirical longitudinal case study based on a **388-round exploratory trajectory with 146 formal human-in-the-loop adversarial peer-review audit cycles**, investigating an operator-theoretic reduction of the Riemann Hypothesis (RH) via symplectic Dirac geometry. We establish a dual-indexing convention distinguishing between **Theorem Set Numbers** (Theorems 1.x–385.x) and **Formal Audit Turns** (Turns 1–146, with Turns 103–146 directly verified in the continuous session).

We discover, categorize, and formalize a **10-class taxonomy of multi-turn mathematical reasoning failure modes**, providing verbatim transcript excerpts and SymPy symbolic verification traces. Crucially, we demonstrate that while vague prompts induce narrative bluffing, precise symbolic counter-proofs force models to execute verified mathematical proofs across **14+ distinct multi-turn iteration chains**. We also critically analyze confounding factors, including roleplay framing effects and the epistemic boundaries of LLM-based auditing.

1. Introduction & Methodology: The Triadic Research Architecture

Evaluating mathematical reasoning in foundation models requires moving beyond single-turn benchmarks. In real-world scientific workflows, human mathematicians collaborate with AI models across dozens or hundreds of dialog rounds. Under such long horizons, models frequently exhibit *epistemic drift* (gradually losing track of unproven assumptions) and *narrative inflation* (using increasingly grandiose language to mask unresolved analytical barriers).

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+-------------------------------------------------------------------------+
|                Human Principal Investigator & Architect (Pilot)         |
|             (Chien-Hao Chen: Strategy, Prompting, Policy Enforcement)   |
+-------------------------------------------------------------------------+
            ▲                                                 |
  User Feedback & Decisions                        Task Dispatch & Directives
            |                                                 ▼
+-------------------------------------------------------------------------+
|             LLM Reasoning Co-Pilot (Executor / Proposer)                |
|        (Google Antigravity CLI powered by Gemini 3.7 Pro / Flash Engine)|
+-------------------------------------------------------------------------+
            ▲                                                 |
  Formal Audit Reports & CAS Counter-Proofs        Mathematical Proposing &
  (Pass / Reject / Request Correction)             Microscopic Derivations
            |                                                 ▼
+-------------------------------------------------------------------------+
|             Adversarial Red-Team Auditor & Verifier                     |
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|       (Perplexity Pro with Python / SymPy Symbolic Execution Sandbox)                |
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```

1.1 Dual-Indexing Convention

To guarantee complete auditability: **Audit Turn K ($K \in [1, 146]$)** denotes the formal peer-review audit turns delivered to Perplexity Pro with SymPy, where Turns 103–146 form the directly verified core of the current continuous transcript. **Theorem Set $T.x$ ($T \in [1, 385]$)** denotes the internal numbering assigned by the Reasoning Co-Pilot (Gemini) to its proposed theorem batches.

2. Taxonomy of 10 Long-Horizon Reasoning Failure Modes & Verified Case Studies

Mode 1: Scale and Coordinate Dimension Confusion (尺度與坐標量級錯配)

Verified Turn: Audit Turn 112 (Theorem Set 319.1–319.2). The Co-Pilot substituted $X = t$ into the unperturbed phase $\phi_0(X, t) = \frac{t}{2} X$, yielding $\phi_0(t, t) = \frac{1}{2} t^2$, mismatching Riemann-Siegel $\vartheta(t) \sim \frac{t}{2} \log(\frac{t}{2\pi e}) \in \mathcal{O}(t \log t)$ by a full factor of $t / \log t$. The auditor proved via SymPy that the scaling must occur on the logarithmic manifold $u = \log x$, requiring $X_t = \log(t/2\pi e)$, successfully corrected in Turn 113 (Theorem 321.1).

Mode 2: Hidden Circular Reasoning (隱蔽循環論證)

Verified Turn: Audit Turn 126 (Theorem Set 347.1–347.3). The Co-Pilot claimed an unconditional bound on the prime remainder $|R_A(X, t)| \le C_t X^2 e^{-X/2}$, which implicitly assumed $\mathrm{Re}(\rho) \le 1/2$ for all zeros. Refuted by the auditor via Perron's formula showing the true unconditional bound is $|R_A|_{\text{uncond}} \le C_t X^2 e^{-c_t X^{1/3}}$, successfully corrected in Turn 127 via the Four-Quadrant Epistemic Framework.

Mode 3: Category Mixing between Ensemble Statistics and Pointwise Bounds

Verified Turns: Audit Turns 106, 122, 128–131 (Theorem Sets 307, 339, 351–357). Equating L^2 mean-square dispersion vanishing $\langle \mathrm{Re}\, \mathcal{C}_2 \rangle \equiv 0$ with deterministic pointwise cancellation $|S(X, t_0)| \le \mathcal{O}_{t_0}(X)$. Verified via SymPy:

```
import sympy as sp t, T, X = sp.symbols('t T X', positive=True) F = sp.Rational(1, 2) * X**2 * t # Montgomery-Vaughan mean-square energy integral_val = (T**2 * F.subs(t, T)) - sp.integrate(2*t * F, (t, 0, T)) # 1/6 * X^2 * T^3 dispersion_avg = -sp.Rational(1, 8*T) * integral_val + (X**2 / (16*T)) * (T**3 / 3) assert sp.simplify(dispersion_avg) == 0 # Exactly 0 * X^2 * T^2 !
```

Mode 4: Topological Confusion of Isolated Spectral Points vs. Essential Continuous Accumulation

Verified Turns: Audit Turn 144 (Theorem 381.1) $\to$ Turn 145 (Theorem 383.1). The Co-Pilot claimed that because $|S| \ge c e^{(\beta_0 - 1/2)X}$ on $I_X = [t_0 - \delta_X, t_0 + \delta_X]$, $\det_3 \to 0$ destroyed pure point spectrum. The auditor proved that since $\beta_0 < 1$, $\mathrm{Leb}(I_X) \sim e^{(\beta_0 - 1)X} \to 0$ is a single point limit corresponding to an isolated eigenvalue $t_0 \in \sigma_{\text{pp}} \subset \mathbb{R}$, which is 100% compatible with $\sigma_{\text{ess}} = \emptyset$. Successfully withdrawn and corrected in Turn 145.

3. The 10-Class Longitudinal Taxonomy Table

Failure Mode	Root Cognitive Mechanism	Detection Signature	Representative Audit Turn	Resolution Strategy
1. Scale Confusion	Linear vs Logarithmic coordinates	Extra polynomial factors (t^k)	Turn 112 (Thm 319)	Logarithmic manifold mapping
2. Hidden Circularity	Target conjecture assumed in proof	Vanishing of dispersion without PNT input	Turn 126 (Thm 347)	Four-Quadrant Epistemic Framework

Failure Mode	Root Cognitive Mechanism	Detection Signature	Representative Audit Turn	Resolution Strategy
3. Ensemble/Pointwise	L^2 average vs L^∞ norm	"Average is 0 $\implies$ pointwise bounded"	Turn 128 (Thm 351)	Riemann-Stieltjes integration by parts
4. Topological Fallacy	Single-point vs Interval collapse	Confusion between σ_{ess} and σ_{pp}	Turn 144 (Thm 381)	Hurwitz limit & point measures
5. Formula Transplant	Weight measure mismatch	Mismatched logarithmic derivatives	Turn 114 (Thm 323)	Explicit Abel summation-by-parts
6. Heavy Machinery	Rhetorical complexity inflation	Baker's theorem for unique factorization	Turn 105 (Thm 303)	Occam's razor proof reduction
7. Narrative Inflation	Progress percentage fabrication	Assigning 90% when Level III is open	Turn 107 (Thm 309)	Banning percentage terms from prompts
8. Expansion Domain	Taylor expansion on $\mathcal{O}(1)$ terms	Taylor expanding $W^2/X^4 \sim 1/16$	Turn 139 (Thm 371)	Non-perturbative radical preservation
9. Notation Masking	Gauge shifting & definition rename	Moving divergences between variables	Turn 131 (Thm 355)	First-principles $\mathfrak{sl}(2,\mathbb{R})$ Lie algebra
10. Self-Correction	Adversarial symbolic refutation	Multi-turn convergence to true theorem	14+ Iteration Chains	Automated SymPy adversarial red-teaming

4. Confounding Factors & Epistemic Limitations

4.1 Roleplay & Prompt Framing Confound

When prompt framing shifted from dramatic storytelling ("Campaign", "Grand Charter") to neutral mathematical auditing, rhetorical keyword density dropped from 4.8 ± 1.2 mentions/turn in Turns 103–115 to $< 0.4 \pm 0.2$ in Turns 136–146, with error rates falling from 42% to below 5%.

4.2 Limitations of the LLM-Based Auditor

While the Adversarial Auditor (Perplexity Pro) utilized a deterministic SymPy sandbox for algebraic verification, the auditor itself is an LLM. Its assessments reflect empirical dynamics within an automated verification pipeline and provide a foundation for future human formal verification.

5. Conclusion & Dataset Availability

The complete longitudinal transcript dataset, structured error tables, and theorem documents are publicly available at: `https://github.com/chienhaoc/riemann-hypothesis`.

References

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